# Jianxuan Shao

Sydney, Australia | jsha0336@uni.sydney.edu.au | +61 422 190 236 | Portfolio

## Research Interests

I aim to build personal AI that follows ongoing activities, reasons and acts with users, and learns from repeated collaboration. I am interested in combining dialogue and task history with screen, audio and visual context. My preparation spans architecture, immersive systems, interactive prototyping and machine-learning evaluation.

## Education

**University of Sydney** **Feb 2025 – Present**

Master of Computer Science | Focus: Data Science and AI | Expected Dec 2026

Semester 1, 2026 average: 84.5/100; average across 12 graded units: 79.33/100. High Distinctions: Introduction to Cybersecurity 92; Machine Learning and Data Mining 89; Introduction to Programming 87; Computer and Network Organisation 85.

**Hebei University** **Sep 2018 – Jun 2023**

Bachelor of Engineering in Architecture | GPA 4.05/5.0 | Average 86.0/100

## Human AI Collaboration

### Boundaries

2026 | Independent browser prototype | AI-assisted JavaScript and Three.js | Project page

- Built an interactive 3D exhibition layout to explore how an assistant can offer useful moves while respecting the parts of a workspace that a user wants to keep in place.
- Separated the assistant’s proposal from the current layout with a translucent preview, borrowing the idea of placing tracing paper over a drawing. People can inspect a move, apply it and undo it.
- Implemented rules that specify an object or area, the affected action and how long the protection lasts. The prototype checks each scripted candidate against the active rule before offering a preview. Stage-specific rules expire when the task moves on.
- Logged offers, blocked proposals, confirmations and undo actions. The browser demo can export the session log so a reviewer can trace how the same suggestion is handled under different rules.
- Delivered a browser prototype with scripted candidate moves. User studies, speech or gesture input and testing in a headset are proposed extensions.

### Game Fellow

2025–present | Independent prototype and supervised research development | Python, Pygame | Project page

- Implemented a second ship that maintains formation, intercepts threats and provides cover fire. Used bounded movement, stable behaviour states and cooldowns to limit rapid switching and repeated interruptions.
- Explored a separate screen-based route in which a vision model interpreted game screenshots and informed companion actions and speech. The public demo uses local game-state rules and can be played without a vision model.
- Prepared a counterbalanced comparison of Solo and Fellow play with paired random seeds and event logs. Drafted questionnaires about teammate presence, trust, usefulness and player control. Participant evaluation is planned.

## Machine Learning and Model Evaluation

### Explainable AI Generated Image Detection

2026 | COMP5703 MSc capstone, eight-person team | Baseline and error analysis | Project page

- Co-developed the evaluation questions and implemented the direct-detection baseline. Audited 60,000 predictions from ten runs against the canonical image manifest and checked parsed labels and prediction alignment.
- The team compared two Qwen3.5-9B routes, one starting from a verdict and one from structured visual evidence, with a ResNet-50 visual-forensics branch. A logistic fusion model combined the branch outputs. Evaluation used 6,000 FakeBench images across ten generators.
- Independently reconstructed all 900 frozen fusion decisions from saved branch outputs and coefficients. All labels matched without retraining or changing the decision threshold.

## User Research and Immersive Design

### Damo Cave Light and Shadow Art Museum

2023 installation | Interactive Media Intern, Shanghai Fengyuzhu | Jul 2023–Jul 2024 | Project page

- Helped set up and debug the installation on site. The team system brought together multi-channel projection blending, wall-touch controls and radar sensing, so integration involved checking display and interaction together in the space.
- Built a later AI-assisted web exploration using MediaPipe and Canvas to map hand and body motion to scene brightness, opening effects and floor responses. Added pointer controls so visitors can try the interaction without enabling a camera.

### Signal Box

2026 | IDEA9106 interaction design coursework | Interviews, prototyping and usability testing | Project page

- Built a physical box, printed waiting slips and story cards, and developed an audio website for instructions and the card reveal. The prototype presented another applicant’s experience in response to the situation on the slip.
- Evaluated the first prototype in four exploratory think-aloud sessions. A facilitator selected a matching card behind the box as a Wizard of Oz test. For the session I reviewed, I recorded timing, hesitation, prompts and whether the reader found the story source.
- Built Lucky Box to demonstrate the revised flow. Added light and sound after testers hesitated or pressed again. After readers needed help finding the source, I added a visible prompt on the card and gave the QR code a direct route to the source.
- The revised website demonstrates one fixed example, with the option to check the source or share a waiting experience. Follow-up usability testing is planned.

## Technical and Research Skills

Programming and data: Python, SQL, Pygame; preprocessing, model evaluation and reproducible analysis.

Interaction and spatial design: State-based behaviour, event logging, CAD, Rhino/Grasshopper, 3D modelling, projection and lighting integration.

Research practice: Research-question development, comparative evaluation, study preparation, prototype iteration, error analysis and technical reporting.